

Targeted Deep Survival Contrasts: Valid Inference for Treatment-Specific Survival Benefit with Neural Networks

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Abstract

Clinical deployment of neural survival models increasingly requires *counterfactual* claims—how much a treatment would change survival in a population—not just prognostic risk scores. Such claims demand valid inference for treatment-specific survival contrasts under confounding and covariate-dependent censoring, targets for which standard deep survival estimators are biased and provide no honest uncertainty. Building on Targeted Deep Architectures (TDA), which embeds TMLE in a network’s weight space, we target the full vector of treatment-specific survival curves $\{S_1(t), S_0(t)\}$ over a time grid, and hence the benefit curve $S_1(t) - S_0(t)$ and the restricted mean survival time (RMST) difference. A single universal targeting path—a ridge projection of the stacked efficient influence functions onto closed-form last-layer gradients—simultaneously solves the projected estimating equations for all coordinates within the neural working submodel, a one-step residual top-up restores the unrestricted target, and a multiplier bootstrap yields simultaneous confidence bands for the benefit curve over a prespecified time grid. In 200-replication simulations with confounded treatment, sign-varying effect heterogeneity, and censoring that depends on treatment and covariates, our estimator attains nominal pointwise coverage (95.6%) and near-nominal simultaneous band coverage (94.0%), with benefit-curve bias roughly a third of—and intervals 8% narrower than—a per-timepoint one-step (AIPCW) correction built from the same nuisance fits at comparable coverage; a head-to-head mechanism study shows both of TDSC’s ingredients matter, with MSE improving from one-step to output-space survival TMLE to TDSC on identical nuisances, an isotonized one-step closing essentially none of the gap, and causal survival forests carrying visible regularization bias into this population-level target. We provide formal guarantees—joint asymptotic linearity across all coordinates, simultaneous-band validity, and a residual “top-up” that provably restores double robustness—for a cross-fitted variant requiring no Donsker conditions, which is empirically nominal at a quantified efficiency cost (about a third wider intervals). Misspecification and small-sample experiments delineate a three-tier operating envelope: weight-space targeting delivers the best finite-sample efficiency in the well-specified scenarios considered and is the most exposed when the initial fit fails, with the output-space TMLE intermediate and the one-step the reliable fallback.

1 Introduction

Neural networks are now standard tools for individual-level survival prediction from rich clinical inputs, including digital pathology and other high-dimensional modalities. Clinical decisions, however, hinge on *contrasts*: how survival would differ under one treatment versus another. Estimating such contrasts from observational data raises two familiar obstacles—confounding and censoring that depends on treatment and covariates—and a less appreciated third: a network trained for predictive loss is a biased, inefficient estimator of any low-dimensional causal functional [3, 10], and its naive standard errors are invalid.

Semiparametric theory offers the remedy: debias a flexible initial fit by solving the efficient influence function (EIF) equation, via one-step/AIPCW corrections [3, 6] or targeted maximum likelihood estimation (TMLE) [12]. For survival targets these tools are mature [8, 1, 9], but standard implementations correct *outside* the model: the one-step adds the empirical mean of the EIF to a plug-in, and classical TMLE fluctuates the output distribution after training. Both become awkward when the target is a whole *vector*—two survival curves on a fine grid—where per-coordinate corrections need not respect monotonicity and can push estimates off the model space. Targeted Deep Architectures (TDA) [7] instead embeds the TMLE fluctuation *inside* the network: a small targeting subset of weights is updated along the projection of the EIF onto per-sample loss gradients, so the fitted network itself is the debiased estimator. TDA was demonstrated for the average treatment effect and the marginal survival curve without treatment; the clinically decisive object—the causal contrast of treatment-specific survival curves under confounding and dependent censoring—was not addressed.

Related work. Neural estimators of treatment-specific survival curves exist—notably SurvITE [5], which targets exactly our estimand with balanced discrete-time hazard networks, and counterfactual time-to-event models [2]—but they provide point estimates without valid confidence statements; that is the gap we fill. Conversely, survival TMLEs with simultaneous inference exist [1, 9], including universal one-dimensional least-favorable submodels that target an entire survival curve at once; but those universal paths are constructed *analytically*, per estimand and likelihood. TDSC’s contribution on this axis is not that a universal path exists—it is that the path is obtained *generically* from the architecture’s score gradients, with no estimand-specific analytic derivation, for any network and any stack of pathwise-differentiable targets. Causal survival forests [4] give honest inference for conditional effects at fixed horizons rather than the full population benefit curve.

Contributions. We propose *Targeted Deep Survival Contrasts* (TDSC): (1) the target vector $\Psi = (S_1(t_1), \dots, S_1(t_K), S_0(t_1), \dots, S_0(t_K))$ with its stacked EIFs combining treatment and censoring weights (Section 2); (2) closed-form per-sample last-layer gradients making the universal targeting update for all $2K$ coordinates one ridge regression per iteration (Algorithm 1), with multiplier-bootstrap sup- t bands for the benefit curve (Section 3); (3) formal guarantees—joint asymptotic linearity, band validity, and a residual top-up that provably restores point-estimate double robustness—for a cross-fitted variant satisfying the sample-splitting premise of the underlying ADML theory, together with a design rule (and documented failure mode) for where the high-dimensional targeting step may be solved (Sections 3–4); (4) a comprehensive simulation program—a 200-replication main study, nuisance-misspecification and sample-size grids, a cross-fitting comparison, and head-to-head external comparators (output-space survival TMLE, causal survival forests)—that decomposes the mechanism behind TDSC’s gains and maps its operating envelope (Section 5).

2 Setup, estimands, and efficient influence functions

Data. We observe n i.i.d. copies of $O = (X, A, \tilde{T}, \Delta)$: covariates $X \in \mathbb{R}^d$, binary treatment $A \in \{0, 1\}$, discrete follow-up time $\tilde{T} \in \{1, \dots, K\}$, event indicator Δ . Time is coarsened into K bins; within a bin, events are recorded before censoring, and subjects surviving bin K are administratively censored. Let $Y(k) = \mathbf{1}\{\tilde{T} \geq k\}$ (at risk) and $dN(k) = \mathbf{1}\{\tilde{T} = k, \Delta = 1\}$ (event), and write $P_n f = n^{-1} \sum_i f(O_i)$. The likelihood factorizes through the propensity $g(X) = P(A = 1 \mid X)$, the event hazard $h(k \mid a, x)$, and the censoring hazard $h_c(k \mid a, x)$, with conditional survival $S(t \mid a, x) = \prod_{k \leq t} \{1 - h(k \mid a, x)\}$ and censoring survival S_c defined analogously.

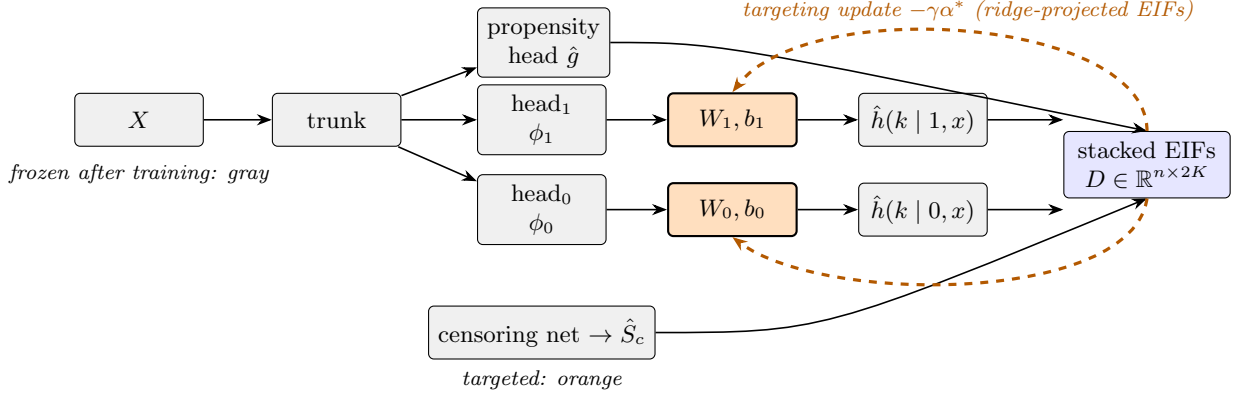


Figure 1: TDSC pipeline. A DragonNet-style survival network produces per-arm hazards; the propensity head and a separate censoring network supply the weights inside the stacked EIFs. Only the final linear layers (W_a, b_a) are updated, along the universal direction obtained by ridge-projecting all $2K$ EIFs onto their closed-form per-sample gradients; everything else stays frozen.

Estimands. Under consistency, no unmeasured confounding, coarsening-at-random censoring (censoring may depend on A and X but not further on the event time), and positivity for treatment and censoring, the treatment-specific curves $S_a(t) = \mathbb{E}_X[S(t | a, X)]$, $a \in \{0, 1\}$, $t \in \{1, \dots, K\}$, are identified. Our target is the $2K$ -vector $\Psi = (S_1(\cdot), S_0(\cdot))$; the *benefit curve* $\beta(t) = S_1(t) - S_0(t)$ and the RMST difference $\Delta_{\text{RMST}} = \sum_{t=1}^K \beta(t)$ (bin width one; the contrast in expected bins survived through horizon K) are linear in Ψ , so their influence functions follow by linearity.

Efficient influence functions. For each coordinate $S_a(t)$ the EIF at P is the discrete-time analogue of Moore and van der Laan [8]:

$$D_{a,t}(O) = - \sum_{k=1}^t \frac{\mathbf{1}\{A = a\}}{g_a(X) S_c(k-1 | a, X)} \frac{S(t | a, X)}{S(k | a, X)} \left(dN(k) - Y(k) h(k | a, X) \right) + S(t | a, X) - S_a(t), \quad (1)$$

with $g_1 = g$, $g_0 = 1 - g$. The first term is a weighted sum of martingale residuals—the within-bin “surprise” of an event relative to the fitted hazard—inversely weighted by the probability of receiving arm a and remaining uncensored through bin $k - 1$; the second is the plug-in centering term. Stacking (1) over a and t gives the $n \times 2K$ influence matrix D whose column means $P_n D_{a,t}$ are the first-order bias terms; the restricted targeting step drives their *projections* onto the working score span to zero, and the residual top-up removes what remains.

3 Targeted Deep Survival Contrasts

Architecture and initial fit. We use a DragonNet-style network [10] adapted to discrete-time survival: a shared trunk feeding a propensity head for g and two outcome heads producing hazard logits $\{h(k | a, x)\}_{k=1}^K$ for $a \in \{0, 1\}$, trained on the observed-arm discrete-time negative log-likelihood (masked per-bin binary cross-entropy) plus a propensity term, with early stopping. A separate network fits the censoring hazard. Any backbone—including a frozen foundation-model encoder with a trained head—can replace the trunk. Full architecture and hyperparameters are in Appendix A.

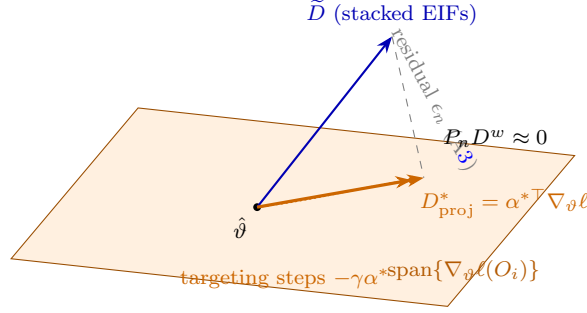


Figure 2: Geometry of the targeting update. The stacked nonparametric EIFs $\tilde{D}$ are projected onto the span of the per-sample loss gradients of the targeting parameters; TDSC steps ϑ along the projected direction until the *projected* estimating equations hold. The observable residual ϵ_n (Assumption 3) measures how much of the nonparametric EIF lies outside the working score span—the gap between the two influence-function geometries and the potential efficiency gain from the restriction, not the parameter-value bias, which the oracle-bias conditions govern separately—and the full-EIF means it leaves behind are exactly what the one-step residual top-up adds back.

Targeting submodel and closed-form gradients. Following TDA, the targeting parameters ϑ are the final linear layers of the two outcome heads, $p = 2K(F+1)$ parameters for head width F . Because those layers are linear in the head features $\phi_a(X)$ and the loss is a per-bin Bernoulli likelihood, the per-sample gradient is available in closed form:

$$\frac{\partial \ell_i}{\partial W_a[k, \cdot]} = \mathbf{1}\{A_i = a\} Y_i(k) (h(k | a, X_i) - dN_i(k)) \phi_a(X_i), \quad (2)$$

and analogously for biases, giving the $n \times p$ score matrix G in one vectorized pass with no per-sample automatic differentiation.

Universal targeting update. Each iteration solves one ridge system for all $2K$ targets, $\alpha = (G^\top G + \lambda I)^{-1} G^\top D$ (with λ scaled to the mean diagonal of $G^\top G$; note $p > n$ here, so the ridge penalty is statistically active), giving per-target directions α_j and projected bias terms $d_j = P_n[G\alpha_j]$. Writing $D^w = G\alpha$ for the resulting *working-submodel* influence functions, the universal direction is $\alpha^* = \sum_j (d_j / \|d\|_2) \alpha_j$ [7, Sec. 2.3], applied as $\vartheta \leftarrow \vartheta - \gamma \alpha^*$ with a backtracking line search on $\sum_j (P_n D_j^w)^2$ (α held fixed within the search). After each accepted step the hazards, influence matrix, and projection are recomputed (g and S_c stay fixed). *The stopping criterion is the projected one:* iteration ends when every $|P_n D_j^w| \leq \widehat{\text{sd}}(D_j^w) / (\sqrt{n} \log n)$ —the estimating equations of the working submodel, which the restricted parameterization can actually solve. The *full*-EIF means $P_n D_j$ need not vanish inside a restricted submodel; they are tracked as a diagnostic, and adding them back as a one-step residual top-up, $\hat{\Psi} + P_n D$, converts the plug-in into an estimator of the unrestricted target (Proposition 3). The second diagnostic is the relative projection residual $\|D - D^w\| / \|D\|$, whose interpretation Section 4 makes precise: it measures how much of the nonparametric EIF lies outside the working score span—the gap between the two influence-function *geometries*, hence the potential efficiency gain from the restriction—not the parameter-value bias $\Psi_n(P_0) - \Psi(P_0)$, which is governed separately by the oracle-bias conditions. The plug-in estimator is, by construction, a pair of monotone survival curves.

Cross-fitted variant. To satisfy the sample-splitting premise of the theory below without Donsker conditions on the network class, TDSC admits a V -fold cross-fitted form: for each fold, the nuisance

Algorithm 1 TDSC universal targeting (no-split form)

Require: trained hazard/propensity network and censoring network; targeting parameters ϑ (final layers of the outcome heads); ridge scale λ

- 1: fix weights $\hat{g}, \hat{S}_c$; compute hazards $\hat{h}_a$, survival curves, and stacked full EIFs $D \in \mathbb{R}^{n \times 2K}$ via (1)
- 2: **loop**
- 3: $G \leftarrow$ per-sample loss gradients w.r.t. ϑ $\triangleright$ closed form (2); no autograd
- 4: $\alpha \leftarrow (G^\top G + \lambda I)^{-1} G^\top D$; $D^w \leftarrow G\alpha$ $\triangleright$ working-submodel EIFs
- 5: **if** $|P_n D_j^w| \leq \widehat{\text{sd}}(D_j^w)/(\sqrt{n} \log n)$ for all j **then break** $\triangleright$ projected criterion
- 6: $d_j \leftarrow P_n D_j^w$; $\alpha^* \leftarrow \sum_j (d_j / \|d\|_2) \alpha_j$ $\triangleright$ universal direction
- 7: backtracking line search: $\vartheta \leftarrow \vartheta - \gamma \alpha^*$, accepting only if $\sum_j (P_n D_j^w)^2$ falls (α fixed within the search); **break** if no γ improves
- 8: recompute $\hat{h}_a$, curves, and D at the updated ϑ
- 9: **end loop**
- 10: **return** plug-ins $\hat{\Psi}_j$; full-residual top-up $\hat{\Psi}_j + P_n D_j$; plug-in SEs $\widehat{\text{sd}}(D_j^w)/\sqrt{n}$, top-up SEs $\widehat{\text{sd}}(D_j)/\sqrt{n}$; multiplier-bootstrap band; diagnostics ($\|D - D^w\|/\|D\|$, $\max_j |P_n D_j|$)

networks are trained *and targeted* on the complement; the held-out fold then supplies evaluation—plug-in curves, influence functions at the targeted fit, and an exact residual correction $P_{n,v}[D]$ so the EIF equations hold exactly on held-out data (by Proposition 3, a cross-fitted one-step at the targeted nuisances). Fold contributions are averaged; the pooled per-observation influence matrix supplies variance and bands.

Why not target fold-wise on the held-out folds? A tempting alternative solves the EIF equations with the full targeting submodel *separately on each held-out fold*. This fold-wise high-dimensional targeting fails badly here, and the mechanism is instructive. It is *not* what classical CV-TMLE [11] does: there, a single *low-dimensional* fluctuation is fit by pooling the cross-validated loss over all validation folds— $O(1)$ parameters against n observations—which is exactly why it is safe (the faithful CV-TMLE analogue here would pool the $2K$ one-dimensional fluctuations of Section 5’s output-space TMLE across folds, which we have not run). TDA’s submodel, by contrast, is the network’s final layers ($p \approx 4,000$ here), so on a fold of $n/V \approx 200$ observations the gradient matrix has rank at most 200 and the fold’s empirical EIF equations can be solved nearly exactly—by fitting the fold’s *sampling noise*, whose order $\text{sd}(D)/\sqrt{n/V}$ rivals the genuine first-order bias the step is meant to remove. The resulting hazard perturbations pass through the nonlinear map $h \mapsto \prod_k (1 - h(k))$, rectifying symmetric fold-level fluctuation into systematic bias, and the fold-fitted targeting layers reintroduce exactly the empirical-process term $(P_n - P_0)(\hat{D} - D_0)$ that cross-fitting is meant to eliminate—now concentrated on the smallest split. Empirically this variant gave 14-fold larger benefit-curve bias than the no-split estimator, 70.5% pointwise and 42% simultaneous coverage (100 replications; store preserved in the repository). The design rule: the dimension of the data-adaptive targeting step must be small relative to the split on which its equations are solved, so targeting belongs with training on the large split, and the held-out fold should receive only evaluation and the linear residual correction. Section 5 compares the no-split and cross-fitted forms empirically.

Inference and simultaneous bands. Two variance prescriptions correspond to the two inferential targets of Section 4: the *top-up* estimator (unrestricted target) uses the full-EIF standard errors $\widehat{\text{sd}}(D_j)/\sqrt{n}$, while the *plug-in* (working target) uses the working-submodel standard errors

$\widehat{\text{sd}}(D_j^w)/\sqrt{n}$, both at the final fit; contrasts inherit influence functions by linearity. For the benefit curve we report a sup- t band over the prespecified grid: with i.i.d. standard normal multipliers ξ_i and centered influence values $\bar{D}_{\beta,t} = P_n D_{\beta,t}$, the 95% quantile c^* of $\max_t |n^{-1} \sum_i \xi_i \{D_{\beta,t}(O_i) - \bar{D}_{\beta,t}\}| / \widehat{\text{se}}_t$ over bootstrap draws gives $\hat{\beta}(t) \pm c^* \widehat{\text{se}}_t$ (c^* floored at 1.96, a conservative choice).

4 Theory

TDSC is TDA applied to a $2K$ -dimensional pathwise-differentiable target, and its guarantees specialize those of Li et al. [7, Thms. 3.1–3.2] and adaptive debiased machine learning (ADML) [13]. We state the specialization for the *cross-fitted* implementation (Section 3; V -fold, nuisances trained and targeted out-of-fold, evaluation with exact residual correction on the held-out fold), which satisfies the sample-splitting premise of those results without Donsker conditions on the network class.

Three influence-function objects must be kept distinct. D_0^{np} denotes the $2K$ -vector of *non-parametric* EIFs (1) at P_0 —the influence function of the unrestricted target $\Psi(P_0)$. $M_n \subseteq \mathcal{M}$ is the data-adaptive working submodel induced by partially unfreezing ϑ , with working parameter $\Psi_n(P_0) = \Psi(\Pi_n P_0)$ and *working-model* EIF D_n^w , the (unpenalized) $L^2(P_0)$ -projection of D_0^{np} onto the closure of the working score span. What the algorithm computes is the ridge-regularized empirical counterpart, $\widehat{D}^{w,\lambda} = G\widehat{\alpha}_\lambda$ with $\widehat{\alpha}_\lambda = (G^\top G + \lambda I)^{-1} G^\top D$; wherever $\widehat{D}^{w,\lambda}$ stands in for D_n^w (the stopping rule, the plug-in standard errors) we assume $\|\widehat{D}^{w,\lambda} - D_n^w\|_{P_0} = o_p(1)$, for which it suffices that the *relative* ridge vanish, $\lambda_n/\text{mean diag}(G^\top G) \rightarrow 0$; our simulations fix that ratio at 0.01 as a finite-sample stabilizer, and its sensitivity is examined in Section 5. Under ADML there is additionally an oracle submodel with EIF D^O to which D_n^w converges; the estimators below are asymptotically linear with D_0^{np} (top-up) or D^O (plug-in), and conflating the two is what a careless reading invites—the entire variance story lives in the gap between them.

Assumption 1 (Identification and positivity). *Consistency, no unmeasured confounding, coarsening-at-random censoring; and $\delta \leq g_0(X) \leq 1 - \delta$, $S_{c,0}(K - 1 \mid a, X) \geq \delta$ a.s. for some $\delta > 0$.*

Assumption 2 (Nuisance rates). *For $f = f(k, a, x)$ write $\|f\|_{P_0}^2 = \sum_a \sum_{k=1}^K \mathbb{E}_{P_0} f(k, a, X)^2$. The cross-fitted weight estimators and the targeted per-fold hazard estimators are each consistent in $\|\cdot\|_{P_0}$, respect the bounds of Assumption 1, and satisfy the product rate $\|\hat{h} - h_0\|_{P_0} (\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}) = o_p(n^{-1/2})$ (a sufficient condition for the sharper per- (a, t) sums of per-bin products appearing in the remainder).*

Assumption 3 (Gradient coverage—plug-in story only). *The projection residual $\epsilon_{n,j} = D_{0,j}^{\text{np}} - D_{n,j}^w$ stabilizes: D_n^w converges in $L^2(P_0)$ to an oracle-model EIF D^O . This assumption is not required for Theorem 1. It governs only the plug-in story (Proposition 1), and the size of $\|\epsilon_n\|$ is a dial, not a defect—but note carefully what it measures: the residual is a statement about influence-function geometry, not parameter values. Even when the oracle submodel contains P_0 (so $\Psi(\Pi_0 P_0) = \Psi(P_0)$ exactly and there is no approximation bias), D^O can remain a strict projection of D_0^{np} onto a smaller tangent space, with $\text{Var}(D^O) < \text{Var}(D_0^{\text{np}})$ —the same parameter with a lower local efficiency bound. That is the ADML superefficiency mechanism. Approximation bias $\Psi_n(P_0) - \Psi(P_0)$ is governed separately by the oracle-bias conditions (C1)–(C2). The empirical residual $\|D - \widehat{D}^{w,\lambda}\|/\|D\|$ estimates the geometric gap.*

Theorem 1 (Joint asymptotic linearity and efficiency of the top-up estimator). *The full-residual top-up estimator is the cross-fitted AIPCW one-step evaluated at the targeted nuisances (Proposition 3); no separate “EIF equations solved” condition is needed. Under Assumptions 1 and 2*

alone (Assumption 3 not required), $\widehat{\Psi}^+$ satisfies jointly in $\mathbb{R}^{2K}$ $\widehat{\Psi}^+ - \Psi(P_0) = P_n D_0^{\text{np}} + o_p(n^{-1/2})$, so $\sqrt{n}(\widehat{\Psi}^+ - \Psi(P_0)) \rightsquigarrow N(0, \Sigma_0)$ with $\Sigma_0 = P_0 D_0^{\text{np}} (D_0^{\text{np}})^\top$: the estimator is regular and attains the nonparametric efficiency bound. Targeting enters only through Assumption 2—it must not degrade, and is designed to improve, the targeted hazards’ rates—so the theorem claims no asymptotic variance advantage for weight-space targeting; any benefit is finite-sample, through better nuisances.

Proposition 1 (Cross-fitted plug-in: working parameter and superefficiency). *The cross-fitted TDSC plug-in (no top-up) is instead an ADML-type estimator of the working parameter $\Psi_n(P_0) = \Psi(\Pi_n P_0)$: under Assumption 3 and the conditions of Li et al. [7, Thms. 3.1–3.2]—including the oracle-bias conditions (C1)–(C2) under which $\Psi_n(P_0)$ may replace $\Psi(P_0)$ —it is asymptotically linear with the oracle EIF D^O , whose variance is no larger than Σ_0 and strictly smaller whenever the projection residual of Assumption 3 does not vanish (superefficiency, with the usual ADML price: nonregularity for the unrestricted target under local alternatives outside the oracle model). Its inference must therefore use the working-model standard errors $\widehat{\text{sd}}(D_j^w)/\sqrt{n}$; pairing the plug-in with full-EIF standard errors is conservative. Both signatures—lower MSE than Theorem 1’s estimator, and over-coverage under full-EIF variance—appear in Section 5.*

Corollary 1 (Contrasts). *The benefit curve $\hat{\beta}(\cdot)$ and $\widehat{\Delta}_{\text{RMST}}$ are asymptotically linear by linearity of both maps, with influence functions $D_{\beta,t}^{\text{np}} = D_{1,t}^{\text{np}} - D_{0,t}^{\text{np}}$ and $\sum_t D_{\beta,t}^{\text{np}}$ for the top-up estimator of Theorem 1, and $D_{\beta,t}^O = D_{1,t}^O - D_{0,t}^O$ and $\sum_t D_{\beta,t}^O$ for the plug-in of Proposition 1.*

Proposition 2 (Simultaneous band). *Under the conditions of Theorem 1, the Gaussian-multiplier estimate of the 95% quantile of $\max_t |\mathbb{G}_t|/\sigma_t$, for $\mathbb{G}$ the limiting Gaussian of Corollary 1, is consistent, and the sup- t band covers the benefit curve over the prespecified grid with asymptotic probability at least 0.95 (the $c^* \geq 1.96$ floor can only enlarge it). Under Proposition 1 plus consistent estimation of $\text{Cov}(D^O)$ by the empirical working-model IFs, the same multiplier construction applies to the plug-in.*

Proposition 3 (Top-up restores double robustness). *Define the top-up estimator with the full residual, $\widehat{\Psi}^+ = \widehat{\Psi} + P_n \widehat{D}$; it equals the AIPCW one-step evaluated at the targeted nuisances. Consequently, under Assumption 1 and convergence of the estimated nuisances to some limits, $\widehat{\Psi}^+$ is consistent for $\Psi(P_0)$ if either the hazard limit equals h_0 or the weight limits equal $(g_0, S_{c,0})$ jointly—regardless of whether targeting converged. (The selective variant implemented in the no-split code, which corrects only coordinates above tolerance, differs from $\widehat{\Psi}^+$ coordinate-wise by at most $\eta_j = \widehat{\text{sd}}(D_j)/(\sqrt{n} \log n) = o_p(n^{-1/2})$, so the conclusion is unaffected.)*

Proof sketches are in Appendix B; they specialize the ADML arguments using the exact second-order remainder of the survival EIF (a telescoping identity), with cross-fitting controlling the empirical-process term.

Remark 1 (No-split implementation and scope). Theorem 1 covers the cross-fitted top-up estimator and Proposition 1 the cross-fitted plug-in; the no-split implementation is covered by neither as stated (it violates the sample-splitting premise of TDA’s first-order expansion), and its validity at moderate n is an empirical matter—Section 5 finds nominal behavior at $n \geq 1000$. Two further caveats. Double robustness of the *point estimate* (Proposition 3) does not extend to inference: consistent variance estimation still requires the relevant nuisances, and Section 5 quantifies the failure modes. Superefficiency likewise cuts both ways: narrower intervals when M_n approximates P_0 well, undercoverage when it does not—the empirical price appears under outcome misspecification and at small n .

5 Simulation study

Design. Covariates $X \sim N(0, I_{10})$; treatment $A \mid X \sim \text{Bern}(\text{expit}(0.9X_1 - 0.7X_2 + 0.6X_3))$, truncated to $[0.05, 0.95]$; event hazard $h(k \mid a, x) = \text{expit}(-3.1 + 2k/K + 0.7x_1 + 0.5x_2 + 0.4x_3 - 0.3x_4 + a\tau(x))$ with $\tau(x) = -1 + 0.8x_1 + 0.6x_3$, so the treatment effect changes sign across the population; censoring hazard $\text{expit}(-3.6 + 0.5x_1 + 0.4x_2 + 0.4a + 0.3x_5)$, dependent on treatment and on covariates shared with the outcome (coarsening at random holds; unadjusted estimators are biased). We use $K = 30$, $n = 1000$ ($\approx 61\%$ events, 47% treated in the demo replication). Ground truth is computed by Monte Carlo integration over 4×10^5 covariate draws (given each draw the conditional survival is evaluated exactly, so the integration error is negligible at the reported precision); the true Δ_{RMST} is 5.45 bins.

Estimators. (i) **Naive plug-in**: the initial network’s curves; (ii) **unadjusted Kaplan–Meier** by arm; (iii) **one-step (AIPCW)**: per-timepoint $\hat{\Psi}_j^{\text{plug}} + P_n D_j$ at the initial fit, using the same nuisances as TDSC; (iv) **TDSC**, plus its residual top-up variant; (v) in the scenario grid below, an IPTW $\times$ IPCW-weighted KM and an *isotonized* one-step (each arm projected onto monotone curves), which separates how much of TDSC’s advantage is explained by monotonicity alone; (vi) a per-timepoint **output-space survival TMLE**—classical TMLE fluctuating the fitted hazards themselves (“output space,” in contrast to TDSC’s weight-space update), in the style of Moore and van der Laan [8]: each of the $2K$ coordinates gets its own one-dimensional logistic hazard fluctuation with clever covariate $H_t(k, x) = -S(t \mid a, x) / \{S(k \mid a, x) g_a(x) S_c(k-1 \mid a, x)\}$, Newton-solved and iterated to convergence, using the *same* nuisance fits as (iii)–(iv); and (vii) **causal survival forests** [4], which fit their own nuisances, evaluated as the doubly robust population benefit at horizons $\{5, 10, \dots, 25\}$.

Results. Table 4 reports time-averaged operating characteristics over 200 Monte Carlo replications (per-cell coverage MC SE ≈ 1.5 points). Four findings. First, adjustment is essential: unadjusted KM’s absolute bias (0.121) is more than 50-fold TDSC’s (0.0021), and its average Δ_{RMST} misses the truth of 5.45 entirely. Second, the naive plug-in carries the regularization-induced bias that debiasing exists to remove (0.021; RMST bias -0.38). Third, on the benefit curve TDSC and the one-step both achieve nominal pointwise coverage (95.6% vs 95.1%), but TDSC does so with roughly a third of the bias (0.0021 vs 0.0057) and 8% narrower intervals (0.140 vs 0.151); the MSE point estimates favor TDSC (0.00149 vs 0.00173), though the *paired* replication-level MSE difference (-0.00024 , MC SE 0.00026) is not resolved at 200 replications. The narrower-at-nominal-coverage combination reflects the working-model efficiency of Section 3; the comparison does not isolate weight-space targeting per se from TMLE-versus-one-step, which the isotonized one-step and output-space TMLE comparators in the full study are designed to decompose. Fourth, the two estimators trade off differently on the RMST functional: TDSC is essentially unbiased (0.007 vs -0.161) but has higher variance, giving the one-step the lower RMST MSE (0.53 vs 0.79); both cover at ≈ 95 – 97% . The simultaneous band covered the whole curve in 94.0% of replications, with $c^* \approx 2.9$. Targeting averaged 9.1 iterations. Figure 4 shows one replication (seed 7; representative—per-replication MAE rankings match the table).

Nuisance misspecification and mechanism. A second study (150 replications per arm; propensity fit by a standalone network throughout so the outcome and weight models can be blinded independently) stresses both arms of double robustness by hiding key covariates from one set of nuisances at a time (Table 3). Three conclusions. *Monotonicity is not the mechanism*: in the

well-specified arm the isotonized one-step is indistinguishable from the raw one-step (MSE 0.00154 vs 0.00155), while in this configuration TDSC’s paired MSE advantage is clearly resolved (-0.00052 , MC SE 0.00009)—the gain comes from targeting, not from projecting curves onto the monotone cone. *Point-estimate double robustness holds, with a caveat:* blinding the outcome network to the main confounder and effect modifier explodes the naive plug-in (bias 0.19; RMST bias -5.8) while the EIF-corrected estimators stay an order of magnitude less biased; blinding the weight models instead leaves both corrected estimators protected through the hazard arm while the weighting-only IPTW $\times$ IPCW KM fails (bias 0.076).

Inference is not doubly robust, and TDSC pays first: under outcome misspecification TDSC’s working-model intervals undercover badly (64.6%; top-up 84.3%) while the one-step—whose wider intervals track the nonparametric variance—stays at 95.0%. A *safeguarded* variant—the top-up point estimate paired with the nonparametric (initial-fit) SE—repairs much of the shortfall (coverage 88.8%) at the one-step’s width, but only the one-step itself remains fully nominal, consistent with the targeted fit drifting along a badly misspecified gradient span: even the raw TDSC estimate paired with the nonparametric SE covers only 72%, so the failure lives in point-estimate variability that no IF-based SE captures. Under weight misspecification all corrected estimators undercover mildly ($\approx 89\%$) through the variance estimate. Diagnostics flag the failure in-sample: the mean number of coordinates left above tolerance rises from 37 (well-specified; nominal coverage regardless) to 54 (outcome-misspecified), and the strict $\text{sd}/(\sqrt{n} \log n)$ tolerance was never met in full in any of the 450 replications—the line search exits first—so we recommend reporting unsolved-coordinate counts and applying the residual top-up as defaults.

Sample-size scaling. Repeating the main (well-specified, Table 4 configuration) study at $n \in \{500, 2000\}$ (Table 2) shows the expected $\sqrt{n}$ behavior from $n = 1000$ up: TDSC’s $\sqrt{n} \times \text{bias}$ is flat (0.067 at $n=1000$, 0.062 at $n=2000$), $n \times \text{MSE}$ falls (1.49 to 1.02, dipping below the one-step’s 1.43—the superefficiency signature), and pointwise and simultaneous coverage hold at 94–97%. At $n = 500$ the ranking *reverses*: TDSC’s bias (0.0140) exceeds the one-step’s (0.0056) and its coverage drops (91.4% pointwise, 85.3% simultaneous) while the one-step stays near nominal (94.8%). This is the misspecification lesson in another guise—an underpowered initial fit, whether from blinded covariates or a small sample, erodes the working-model gains first—so at small n the one-step is again the safer instrument.

Cross-fitting. The cross-fitted estimator of Sections 3–4 ($V = 5$, 100 replications paired with the main study’s datasets) attains nominal inference everywhere—95.5% pointwise, 94.0% simultaneous, 95.0% RMST coverage—so the estimator covered by Theorem 1 works as advertised. Its price at $n = 1000$: intervals 34% wider than no-split TDSC (0.187 vs 0.140) and higher MSE (paired difference $+0.00137$, MC SE 0.00022), reflecting nuisances trained on $4n/5$ and the held-out residual correction, whose sampling noise (mean max residual 0.22) more than doubles MSE relative to the cross-fitted plug-in alone (0.00248 vs 0.00104)—the latter the lowest MSE in this comparison, at mildly conservative coverage (97.8%), the empirical signature of Proposition 1. Since both forms are nominal here, the comparison quantifies what the no-split shortcut saves when it is safe: the Donsker concern is empirically immaterial at this scale, and cross-fitting provides the theorem-backed safe harbor—most valuable precisely where no-split inference failed (misspecification, small n), which we leave as the natural next experiment.

External comparators: decomposing the mechanism. Table 1 and Figure 3 report the output-space TMLE and causal survival forests head-to-head with the one-step and TDSC (identical

Table 1: External comparators, benefit curve ($n = 1000$, $K = 30$). One-step and TDSC rows repeat Table 3 (150 replications); output-space TMLE uses the same nuisance seeds (100 replications); causal survival forests fit their own nuisances (100 replications, evaluated at horizons 5–25, so cells are not directly comparable to the 30-point averages). Restricted to the 100 shared replications, the well-specified ordering is unchanged (one-step 0.00158, output-space TMLE 0.00127, TDSC 0.00109).

Scenario	Estimator	Bias	MSE	Coverage	CI width
well-specified	one-step (AIPCW)	0.0051	0.00155	94.5%	0.144
	output-space TMLE	0.0042	0.00127	92.5%	0.134
	TDSC	0.0036	0.00103	95.7%	0.135
outcome mis.	one-step (AIPCW)	0.0135	0.00345	95.0%	0.197
	output-space TMLE	0.0115	0.00223	89.3%	0.149
	TDSC	0.0119	0.01860	64.6%	0.156
weights mis.	one-step (AIPCW)	0.0114	0.00111	89.4%	0.111
	output-space TMLE	0.0132	0.00130	87.1%	0.110
	TDSC	0.0111	0.00114	89.2%	0.111
(own nuisances)	causal survival forest	0.0183	0.00130	89.2%	0.114

nuisance seeds for (iii), (iv), (vi); 100 replications for the new comparators). In the well-specified arm the three same-nuisance estimators order exactly along the two design axes: one-step (MSE 0.00155) $\rightarrow$ output-space TMLE (0.00127) $\rightarrow$ TDSC (0.00103). Moving from additive correction to TMLE buys roughly half the improvement, and moving the TMLE from output space into weight space buys the other half—so weight-space targeting is not redundant given TMLE-ness, answering the decomposition question the isotonized one-step began. TDSC also has the best coverage of the three (95.7% vs 92.5% for the output-space TMLE, whose per-coordinate fluctuations add small-sample noise). Under *outcome* misspecification the ordering inverts for stability: the output-space TMLE’s one-dimensional fluctuations cannot drift the way the $p \approx 4,000$ -dimensional weight-space update can, so it keeps the lowest MSE (0.00223 vs TDSC’s 0.0186) while only the one-step retains nominal coverage (95.0%)—sharpening the operating envelope: weight-space targeting gives the best finite-sample efficiency in the well-specified scenarios considered and is the most exposed when the initial fit fails, with the output-space TMLE the intermediate-robustness option. Under weights misspecification all three are equivalent (≈ 87 – 89% coverage through the variance estimate). Causal survival forests, solving the harder conditional-effect problem with their own nuisances, carry visible regularization bias into the population benefit (bias 0.018, roughly $5\times$ the well-specified TDSC bias in Table 1; 89.2% coverage)—all three debiased semiparametric estimators dominate on this population-level target. Finally, the correct statement of the practical contrast: universal one-dimensional targeting paths exist in output space too [1, 9]—and we implement one as a comparator—but they are derived analytically per estimand and likelihood, whereas TDSC’s path falls out of the architecture’s score gradients generically; the per-timepoint variant’s $2K$ separate fluctuation sequences illustrate the bookkeeping that analytic construction replaces, not a fundamental limitation of output-space TMLE.

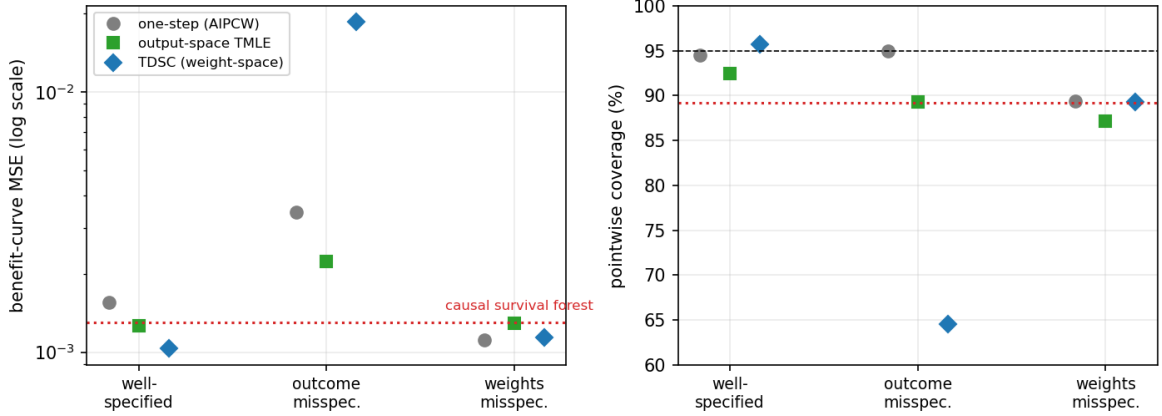


Figure 3: Head-to-head comparison across nuisance scenarios (benefit-curve MSE, log scale; pointwise coverage, 95% dashed). The three same-nuisance estimators order one-step $\rightarrow$ output-space TMLE $\rightarrow$ TDSC when well-specified; under outcome misspecification the weight-space update is the most exposed while the one-step alone stays nominal. The causal survival forest (own nuisances, horizons 5–25) appears as the dotted reference.

Table 2: Sample-size scaling, benefit curve (well-specified, Table 4 configuration; 150/200/150 replications).

n	Estimator	Bias	$n \times \text{MSE}$	Coverage	CI width	Simult. band
500	one-step	0.0056	2.10	94.8%	0.232	—
	TDSC	0.0140	2.77	91.4%	0.198	85.3%
1000	one-step	0.0057	1.73	95.1%	0.151	—
	TDSC	0.0021	1.49	95.6%	0.140	94.0%
2000	one-step	0.0025	1.43	93.7%	0.096	—
	TDSC	0.0014	1.02	96.2%	0.093	96.7%

6 Discussion

TDSC turns a two-headed neural survival network into a valid estimator of population treatment benefit at the cost of a few ridge regressions after training, and the misspecification grid gives the method an honest operating envelope: prefer TDSC when the initial network fits well (narrower intervals at nominal coverage, gains not attributable to monotonicity; $n \gtrsim 1000$ in our design), and fall back when diagnostics (large full-EIF residual means—the top-up magnitudes—or a large projection residual) suggest otherwise or the sample is small—the safeguarded top-up repairs most of the coverage loss, but under gross outcome misspecification, and at $n = 500$, only the one-step is fully reliable. The remaining work: (i) the cross-fitted variant under misspecification and at small n —it is nominal and modestly less efficient in the well-specified case (Section 5), and the settings where no-split inference failed are exactly where its guarantee should earn its keep; (ii) sensitivity to the ridge penalty, targeting-layer choice, and weight-truncation levels (Appendix A); (iii) *calibration-for-benefit*—targeting arm-specific curves within strata of the network’s own predicted benefit, the estimand used to validate individualized-benefit models against randomized trials—and application on frozen pretrained encoders (e.g. pathology foundation models), which the last-layer construction

Table 3: Nuisance-misspecification grid, benefit curve ($n = 1000$, $K = 30$, 150 replications per arm; coverage MC SE $\approx 1.8\text{pp}$). *outcome mis.*: outcome hazards blinded to x_1, x_3 ; *weights mis.*: propensity and censoring models blinded to x_1, x_2 . Propensity is fit by a standalone network (not Table 4’s trunk head), so well-specified cells are not directly comparable to Table 4. The isotonized row changes only the point estimate; the safeguard row changes only the interval (its nonparametric SE equals the one-step’s, hence the equal widths).

Scenario	Estimator	Bias	MSE	Coverage	CI width
well-specified	one-step	0.0051	0.00155	94.5%	0.144
	one-step, isotonized	0.0051	0.00154	—	—
	TDSC	0.0036	0.00103	95.7%	0.135
outcome mis.	naive plug-in	0.1922	0.04110	—	—
	one-step	0.0135	0.00345	95.0%	0.197
	TDSC	0.0119	0.01860	64.6%	0.156
	TDSC + top-up	0.0265	0.01325	84.3%	0.156
	+ nonpar. SE (safeguard)	—	—	88.8%	0.197
weights mis.	IPTW $\times$ IPCW KM	0.0760	0.00720	—	—
	one-step	0.0114	0.00111	89.4%	0.111
	TDSC	0.0111	0.00114	89.2%	0.111

supports directly.

A Reproducibility

All code, raw Monte Carlo stores, and the exact scripts behind every table are at <https://github.com/blind-contours/tdsc>. Networks: trunk $d \rightarrow 64 \rightarrow 64$ (ELU); propensity head $64 \rightarrow 64 \rightarrow 1$; per-arm outcome heads $64 \rightarrow 64 \rightarrow K$; censoring network $(d+1) \rightarrow 64 \rightarrow 64 \rightarrow (K-1)$. Training: full-batch Adam, lr 10^{-3} , max 500 epochs, early stopping (patience 25) on a 20% validation split, propensity loss weight 1.0. Targeting: ridge scale $\lambda = 0.01 \times \text{mean diag}(G^\top G)$ (a finite-sample stabilizer; the theory requires only that the relative ridge vanish, Section 4); $\gamma_0 = 1$ with up to 12 halvings; max 50 iterations; stopping tolerance $\widehat{\text{sd}}(D_j^w)/(\sqrt{n} \log n)$ on the projected means. Truncation: $\hat{g} \in [0.05, 0.95]$, $\hat{S}_c \geq 0.05$, conditional survival floored at 10^{-3} inside (1). Bands: 1000 Gaussian-multiplier draws. Cross-fitting: $V = 5$, fold networks trained and targeted on complements with the same hyperparameters. Output-space TMLE: per-coordinate logistic fluctuations solved by Newton steps (ϵ clipped to $[-5, 5]$), recomputing the clever covariate each round, up to 25 rounds per coordinate, same $\text{sd}/(\sqrt{n} \log n)$ tolerance. Causal survival forests: `grf` 2.6.1, `causal_survival_forest` with `target = "survival.probability"`, 1,000 trees, default tuning, one forest per horizon $\{5, 10, 15, 20, 25\}$, population effect and SE from `average_treatment_effect`. The main tables use the no-split estimator (Table 4 with the trunk propensity head; Tables 3 and 1 with a standalone propensity network). Truth: Monte Carlo integration over 4×10^5 covariate draws.

Table 4: Benefit curve $S_1(t) - S_0(t)$, time-averaged over 30 points ($n = 1000$, $K = 30$; 200 replications, coverage MC SE $\approx 1.5\text{pp}$). Bias is the time-averaged absolute Monte Carlo bias. The paired TDSC–one-step MSE difference is -0.00024 (MC SE 0.00026).

	Bias	MSE	Coverage (ptwise)	CI width
Naive neural plug-in	0.0213	0.0026	—	—
Unadjusted KM	0.1212	0.0163	—	—
One-step (AIPCW)	0.0057	0.0017	95.1%	0.151
TDSC (ours)	0.0021	0.0015	95.6%	0.140
Δ_{RMST} : one-step	bias -0.161	MSE 0.53	96.5%	2.87
Δ_{RMST} : TDSC	bias 0.007	MSE 0.79	95.5%	2.69
TDSC simultaneous band	coverage 94.0% (whole curve)		band width 0.207	

B Proof sketches

Theorem 1. All influence functions in this proof are the nonparametric EIFs (1); superscripts are omitted. Per coordinate, since $\hat{\Psi}_j^+ = \Psi_j(\hat{P}) + P_n \hat{D}_j$ and the EIF has the form $\hat{D}_j = \hat{D}_j^{\text{mart}} + \hat{S}_j - \Psi_j(\hat{P})$, the standard one-step decomposition holds exactly:

$$\hat{\Psi}_j^+ - \Psi_j(P_0) = P_n D_{0,j} + (P_n - P_0)(\hat{D}_j - D_{0,j}) + R_{2,j}, \quad R_{2,j} = P_0 \hat{D}_j + \Psi_j(\hat{P}) - \Psi_j(P_0).$$

For the remainder, the survival EIF is exact to second order: writing $S_h(k) = \prod_{j \leq k} (1 - h(j))$ for the conditional survival under a hazard h , the telescoping identity $S_{\hat{h}}(t) - S_{h_0}(t) = \sum_{k \leq t} S_{h_0}(k - 1)(h_0(k) - \hat{h}(k)) S_{\hat{h}}(t)/S_{\hat{h}}(k)$ gives $R_{2,j}$ as a sum over $k \leq t$ of integrals of $\{\hat{h}(k) - h_0(k)\}$ against $\{\hat{g}\hat{S}_c(k-1) - g_0 S_{c,0}(k-1)\}/\{\hat{g}\hat{S}_c(k-1)\}$ -type differences; under the positivity bounds of Assumption 1, Cauchy–Schwarz gives $|R_{2,j}| \lesssim \|\hat{h} - h_0\|_{P_0} (\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}) = o_p(n^{-1/2})$ by Assumption 2, applied to the targeted per-fold hazards (this display also proves Proposition 3: R_2 vanishes if either factor’s limit is correct). For the empirical-process term, cross-fitting makes $\hat{D}_j$ independent of the fold on which $(P_n - P_0)$ is evaluated, so conditional on the training folds, $(P_n - P_0)(\hat{D}_j - D_{0,j}) = o_p(n^{-1/2})$ whenever $\|\hat{D}_j - D_{0,j}\|_{P_0} = o_p(1)$, which follows from the individual nuisance consistency in Assumption 2 (targeting, which moves ϑ along the score span, is designed to improve rather than degrade those rates). Summing folds yields the coordinate-wise expansion; joint convergence follows from the multivariate CLT applied to the stacked D_0 . Proposition 1 is Li et al. [7, Thms. 3.1–3.2] applied verbatim to the cross-fitted plug-in, with superefficiency from their Theorem 3.2 under (C1)–(C2).

Corollary 1 and Proposition 2. Both maps are bounded linear in Ψ , so asymptotic linearity transfers with the transformed influence functions. Given joint normality with consistently estimated Σ_0 (IF second moments), the conditional multiplier process $n^{-1/2} \sum_i \xi_i \hat{D}_{\beta,t}(O_i)$ converges weakly to the same Gaussian limit (conditional multiplier CLT), so its sup-quantile is consistent for the sup-quantile of the limit; the $c^* \geq 1.96$ floor only enlarges the band.

Proposition 3. By definition $\hat{\Psi}_j^+ = P_n[\hat{S}(t | a, X)] + P_n \hat{D}_j$, which is precisely the AIPCW one-step at nuisances $(\hat{h}^{\text{targ}}, \hat{g}, \hat{S}_c)$. Its population bias is the R_2 term above evaluated at those limits, which vanishes under either-arm consistency. $\square$

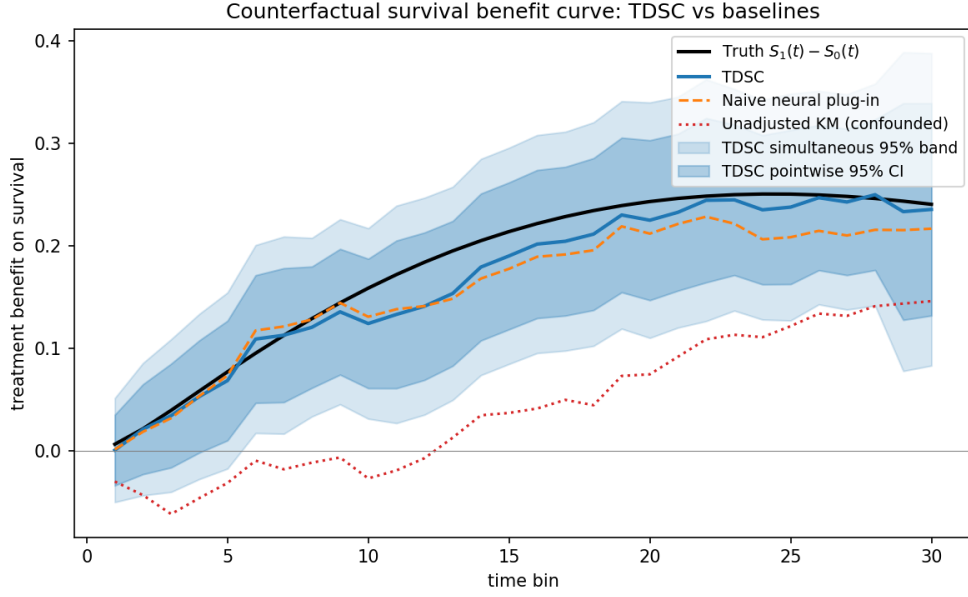


Figure 4: One replication ($n = 1000$, seed 7). The TDSC benefit curve tracks the truth, which lies inside both the pointwise CI and the simultaneous band; the naive plug-in drifts low at later times, and unadjusted KM is severely biased (wrong-signed at early times).

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